

iFocus

Type	Project Type 1 – Image Classifier / Computer Vision
Problem	App-based private drivers suffer high workloads that affect trip's safety
Solution	A computer Vision model to classify driver's exhaustion during a trip

Introduction

Studies suggest that only 10% of Egypt's population own licensed private cars, which explains the high demand on transportation ways. A special type of private transportation, which is application-based private driver (also known as Hail-Riding), faces a rising challenge. The current workloads that drivers endure are too exhausting and may increase the risk of accidents and lower the overall trip experience.

Idea

Facial expressions can talk much about person's attention. We can train a computer vision model to learn 5 facial landmarks for drivers: 2 Eye / 1 Nose tip / 2 Mouth corners. This model can act as a feature inside transportation applications so when a passenger reports a life-threatening trip, the model scans the trip video and gives predictions about the driver's attentiveness.

Data Used

FLD3 – A dataset that contains continuous timeframes of both gender drivers recorded by dashcams during night trips.

Implementation / Technical Decisions

iFocus is built on top of MobileNet to support lightweight operating with low delay. Only a dense layer is connected at the end layer with softmax activation function to give soft predictions for each class: Alert / Yawning / Micro-Sleep

Due to nature of the dataset, Data Augmentation is decided to be on-the-fly to avoid creating & duplicating around 30K images, which will lose the advantage of using MobileNet. Label encoding is preferred to be done using a label map since the data wasn't in a pandas dataframe.

The project folder structure was found better to follow a modular design instead of one central notebook. This can be useful for debugging and future upgrades. Each module serves micro-services that differentiate it from others.